Wenqi Zhao

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Education

Chongqing University, Chongqing, China, M.S. in Mathematics, College of Mathematics and Statistics — Sep 2021 – Jun 2024

- GPA: 3.58/4.00 (85.8/100)
- **Selected coursework:** Image Processing (94), Probability Theory and Mathematical Statistics (97), Scientific Computing (94), Numerical PDEs (90), Differential Equations (88).

Jiangsu Normal University, Xuzhou, China, B.S. in Mathematics, College of Mathematics and Statistics — Sep 2017 – Jun 2021

- GPA: 3.30/4.00 (83/100)
- **Selected coursework:** C Programming (98), Differential Geometry (90), Point-Set Topology (89), Mathematical Statistics (89), Ordinary Differential Equations (88).

Publications

AI Agents for Medicine: A Comprehensive Review of Architectures, Cognitive Modules, and Clinical Workflows (First Author) — 2025

W. Zhao, S. Wang, M. Safari, M. Hu, X. Yang

Medical Image Analysis (submitted)

RefLSM: Linearized Structural-Prior Reflectance Model for Medical Image Segmentation and Bias-Field Correction (First Author) — 2025

W. Zhao, J. Sang, F. Cheng, Y. Shu, D. Li, X. Yang

Pattern Recognition (under review)

Robust image segmentation and bias field correction model based on image structural prior constraint (First Author) — 2024

W. Zhao, J. Sang, Y. Shu, D. Li

Expert Systems with Applications, vol. 251, art. no. 123961

Recent Advances on Multi-Modal Dialogue Systems: A Survey — 2024

F. Cheng, X. Li, H. Wu, J. Sang, ***W. Zhao***

In *Advances in Data Mining and Applications (ADMA 2024)*, LNCS, vol. 15391, Springer, pp. 33–47

Research Experience

AI Agents for Medicine: A Comprehensive Review of Architectures, Cognitive Modules, and Clinical Workflows — Jun 2025 – Present, Atlanta, GA

Research Assistant (Remote), Emory University
Dept. of Computer Science and Informatics and Dept. of Radiation Oncology
Advisor: Prof. Xiaofeng Yang

- **Led** a comprehensive review on medical AI agents, **defining a new taxonomy** for cognitive architectures and mapping components to clinical workflows.

RefLSM: Linearized Structural-Prior Reflectance Model for Medical Image Segmentation — Sep 2024 – Sep 2025, Chongqing, China

Research Assistant, Chongqing University; Co-Advised by Prof. Yonglu Shu (CQU) and Prof. Xiaofeng Yang (Emory)

- **Led** the complete project lifecycle from conceptualization and design to submission, under the co-mentorship of Prof. Shu and Prof. Yang.
- **Developed** RefLSM, a novel variational framework that integrates Retinex decomposition with a **linear structural prior** to segment illumination-invariant reflectance, achieving SOTA on low-contrast medical images.

Robust Image Segmentation and Bias Field Correction Model Based on Image Structural Prior Constraint — Apr 2023 – Mar 2024
Graduate Researcher, Chongqing University; Advisors: Prof. Yonglu Shu, Prof. Dong Li — Chongqing, China

- **Independently conceptualized, designed, and led** the full project lifecycle, resulting in a publication.
- **Proposed** an innovative variational model with spatial structural priors, and **validated** its robustness on diverse medical and synthetic images via extensive comparative experiments (Dice, IoU, JS, VOE).

Recent Advances on Multi-Modal Dialogue Systems: A Survey — Mar 2024 – Sep 2024
Research Collaborator; Advisor: Prof. Xue Li, The University of Queensland — Chongqing, China

- Contributed to a comprehensive review of multi-modal dialogue systems, categorizing them by task type and analyzing benchmark datasets.
- Compared evaluation metrics and identified key challenges, including modality gaps, metric biases, and AI safety.

Teaching Experience

Teaching Assistant — Fundamental Mathematics — Feb 2023 – Jul 2023
Chongqing University — Chongqing, China

- Assisted professors in designing and administering weekly problem sets and examinations for 100+ calculus students, covering topics including limits, derivatives, integrals, and multivariable calculus.

Teaching Assistant — Probability Theory and Mathematical Statistics — Sep 2022 – Mar 2023
Chongqing University — Chongqing, China

- Designed and led tutorial sessions guiding students to apply probability models and statistical inference methods to solve practical problems, leveraging my expertise in stochastic processes and optimization algorithms.

Awards

Academic First-Class Scholarship — Sep 2022 – Jun 2023
Chongqing University

Academic First-Class Scholarship — Sep 2021 – Jun 2022
Chongqing University

3rd Prize, National Mathematics Competition (Category B) — 2019
Chinese Mathematical Society

Skills and Research Interests

Technical Skills: *Matlab, Python, PyTorch, C/C++, Latex*

Research Interests: *AI Agents for Healthcare, Responsible AI, Clinical Decision Support, Medical Computer Vision, Multimodal Machine Learning*